

Article

Dermal Absorption and Quantitative Analysis of Amaranth, a Cosmetic Colorant, in Rat Skin Using an In Vitro Franz Diffusion Model

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Abstract

Amaranth (R2) is used as a color additive in cosmetics. In Korea, R2 is permitted only as a cosmetic colorant and is prohibited in products intended for infants and children under 13 years of age; in Europe, it is regulated solely as a cosmetic colorant rather than a hair dye ingredient. Despite its regulatory relevance, dermal absorption data for R2 are lacking. In this study, percutaneous absorption of R2 was evaluated using the Franz diffusion method with excised rat dorsal skin. Quantitative analysis of R2 was developed and validated using high-performance liquid chromatography (HPLC) in accordance with Korean Ministry of Food and Drug Safety guidelines, demonstrating acceptable linearity ($r^2 = 0.9996$ – 0.9999), accuracy (95.5–104.4%), and precision (0.3–5.8%). Two formulations (skin lotion and cream), each containing 1% R2, were applied at 113 mg/cm² for 24 h. Dermal absorption was assessed by analyzing receptor fluid, skin wash, stratum corneum, epidermis, and dermis. Total dermal absorption of R2 was $3.4 \pm 2.7\%$ for the lotion and 0% for the cream, corresponding to in vitro skin permeabilities of 34.5 ± 27.0 µg/cm² and 0 µg/cm², respectively. Total recovery ranged from $80.3 \pm 8.2\%$ to $91.4 \pm 19.4\%$. These results provide essential data for cosmetic risk assessment of R2.

Keywords: amaranth; Franz diffusion method; dermal absorption; high-performance liquid chromatography (HPLC)



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1. Introduction

Amaranth (R2; C₂₀H₁₁N₂Na₃O₁₀S₃; E123, CAS no. 915-67-3) is a synthetic, water-soluble anionic azo dye widely used as a food colorant and cosmetic ingredient [1–3]. Its physicochemical properties are summarized in Table 1. Azo colorants are characterized by –N=N– linkages connecting aromatic rings, and although they are not inherently carcinogenic, reductive cleavage of the azo bond can release aromatic amines, some of which are recognized carcinogens (Figure 1). Consequently, azo-based colorants are often considered environmental contaminants with potential implications for human health and ecological safety [4–9].

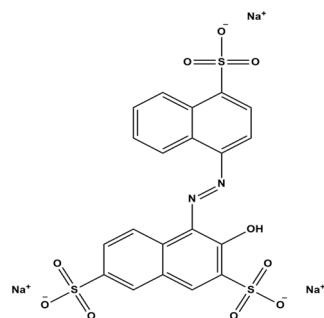


Figure 1. Chemical structure of Amaranth (R2).

In addition, R2 is regulated as a cosmetic colorant in Korea. Under Korea Ministry of Food and Drug Safety (MFDS) standards, R2 is permitted solely as a cosmetic colorant, but it is not allowed in products specifically labeled for infants or for children ≤ 13 years [10]. In the European Union, R2 is listed in Annex IV as an allowed colorant under Regulation (EC) No 1223/2009 [11]; however, its use as a hair-dye substance is prohibited (Annex II, entry 1350). Thus, within the EU it is managed as a cosmetic colorant—not as a hair dye ingredient [12,13].

The cosmetics market has shown steady growth; however, continuous concerns regarding the safety of cosmetics have highlighted the need for further investigation and research. Among cosmetic colorants, issues related to tar pigments have been particularly noteworthy. For example, R2 and New Coccine (R102, CAS no. 2611-82-7) have been prohibited in children's foods to strengthen child safety protections, and their use has also been banned in oral hygiene products, such as mouthwashes and toothpastes, due to the potential for adverse health effects in children. Although no harmful effects have been directly demonstrated in humans, animal studies have raised concerns regarding carcinogenic potential and long-term exposure, leading to restrictions on the use of R102 in the United States since 1976 [14]. In addition, Sunset Yellow FCF (CAS no. 1325-37-7) has been reported to induce renal tumors, urticaria, allergic reactions, and chromosomal damage. Reflecting these safety concerns, the European Food Safety Authority [15] established an acceptable daily intake (ADI) of 4 mg/kg bw/day for this compound when used as a food additive [16]. Despite such regulatory restrictions and toxicological findings for certain synthetic dyes, experimental evidence suggests that not all tar pigments pose the same level of risk. According to the European Chemicals Agency (ECHA) (2025), a rabbit skin-irritation study conducted in accordance with OECD Test Guideline 404 ($n = 9$ per group) using distilled water or an ointment as the vehicle found that topical R2 at 0.1% and 1% was non-irritating and produced no evidence of systemic toxicity [17]. Consistently, ECHA (2025) reported that, in an Organisation for Economic Co-operation and Development (OECD) TG 439 assay using the EpiDerm™ reconstructed human epidermis, undiluted R2 (25 mg, 1 h) yielded a mean tissue viability of 101.7% relative to the Dulbecco's phosphate-buffered saline (DPBS) negative control (30 μ L), whereas the 5% SDS positive control reduced viability to approximately 4.5% of control; negative-control optical density (OD) values ranged from 1.195 to 1.430 after 20–25 DPBS rinses [17].

Thus, the safety of cosmetics is evaluated through risk assessment, which requires comprehensive information on the chemical structure, physicochemical properties, toxicological profiles, and patterns of human exposure to the ingredients [18–20]. Exposure assessment is the process of quantitatively and/or qualitatively estimating the amount or level of a hazardous substance to which humans may be exposed through the use of cosmetics or other products. In this context, the systemic exposure dose (SED) is considered, and the determination of dermal absorption is a critical parameter in calculating SED

values [21–23]. Therefore, the evaluation of dermal absorption is essential for conducting a reliable risk assessment of cosmetic ingredients.

However, to date, no dermal absorption studies on R2 in cosmetic formulations have been reported. Therefore, the present study aimed to evaluate its dermal absorption using an *in vitro* Franz diffusion cell system. The Franz diffusion cell is widely employed for dermal absorption and penetration studies, as it provides a controlled and standardized environment for assessing the permeation of drugs and cosmetic compounds through skin or artificial membranes, thereby generating reliable data for product development [24,25]. It is particularly valuable for formulation optimization, enabling direct comparisons of how different ingredients or vehicles influence permeation and absorption rates [26,27]. In addition, it facilitates comparative studies across multiple formulations or products, assisting researchers in identifying the most effective options. The versatility of this system allows adaptation to different skin types or conditions, making it suitable for evaluating a wide range of formulations, including creams, gels, ointments, and patches [28–30]. Moreover, modern Franz diffusion systems support automation, which increases throughput, improves reproducibility, and reduces human error during sample collection and analysis [31]. Accordingly, the objective of this study was to determine the dermal absorption of R2 as part of its risk assessment using the *in vitro* Franz diffusion cell system. Accordingly, the objective of this study was to determine the dermal absorption of R2 as part of its risk assessment using the *in vitro* Franz diffusion cell system. Due to increasing regulatory and safety concerns associated with synthetic dyes, the development and use of alternative cosmetic colorants are being actively explored. However, quantitative dermal absorption data for Amaranth (R2) remain limited. Therefore, the present study aimed to generate exposure data required for cosmetic risk assessment and to support future decision-making regarding its continued use or potential substitution.

Table 1. Physical and chemical properties of amaranth (R2).

Item	Substance	Ref.
INCI name	CI 16185	[32]
IUPAC name	Trisodium 3-hydroxy-4-(4'-sulphonatonaphthylazo)naphthalene-2,7-disulphonate	[33]
CAS number	915-67-3	
Molecular formula	C ₂₀ H ₁₁ N ₂ Na ₃ O ₁₀ S ₃	
Molecular weight	604.5 g/mol	
Color/form	Dark, reddish-brown powder	[34]
Topological Polar Surface Area	242 Å ²	
Melting point	greater than 300 °C	
Density	0.393 at 20 °C	
Water Solubility	50,000 mg/L (20 °C)	
LogP (log Kow)	−5.13	

Abbreviations: INCI, International Nomenclature of Cosmetic Ingredients; IUPAC, International Union of Pure and Applied Chemistry; CAS, Chemical Abstracts Service.

2. Materials and Methods

2.1. Chemicals

R2 was purchased from Sigma Aldrich Co. (St. Louis, MO, USA). Formic acid was obtained from Merck Millipore Co. (Burlington, MA, USA). Phosphate-buffered saline

(PBS) was purchased from Sigma Aldrich (St. Louis, MO, USA). Distilled water (D.W) and acetonitrile (ACN) were obtained from Honeywell Burdick & Jackson Co. (Muskegon, MI, USA).

2.2. Preparation of Test Formulations

Two different formulations of R2 were prepared and used as test preparations in this study. Skin lotion represents a typical oil-in-water (O/W) emulsion, while cream corresponds to a conventional water-in-oil (W/O) emulsion. The detailed compositions of each formulation are provided in Table 2. Both formulations contained 1.0% R2.

Table 2. Formulations of skin lotion and cream containing 1% amaranth (R2).

Ingredients (% w/w) *	Formulation	
	Skin Lotion	Cream
Distilled water	82.41	49
Glycerin	11.06	-
Ethanol 95%	5.53	-
Lanette wax	-	15
Liquid paraffin	-	12.50
Petrolatum	-	22.50
Amaranth (R2)	1	1
Total	100	100

* % w/w (percent weight by weight). Distilled water—aqueous phase/solvent, Glycerin—humectant, Ethanol 95%—sol-vent/penetration enhancer, Lanette wax—emulsifier/consistency agent, Liquid paraffin—emollient/oil phase, Petrolatum—occlusive agent.

2.3. HPLC Instruments and Conditions

High-performance liquid chromatography (HPLC) analysis of R2 was performed using a Shimadzu system (Shimadzu Corporation, Kyoto, Japan) equipped with a CBM-20A system controller, LC-20AD isocratic pump, SIL-20A autosampler, SPD-20A UV detector (523 nm), and CTO-20A column oven. Chromatographic separation was achieved on a KR100-5C18 column (250 × 4.6 mm, 5 µm; Kromasil, Bohus, Sweden) coupled with a Security Guard Cartridge RP-1 (4 × 3.0 mm; Phenomenex, Torrance, CA, USA). The mobile phase consisted of 90% methanol containing 0.1 M ammonium acetate (pH 6.8) and 10% distilled water, delivered at a flow rate of 1.0 mL/min. The column oven was maintained at 40 °C, and the injection volume for all samples was 20 µL.

2.4. Stability Test

For the stability evaluation, R2 was prepared at a concentration of 10 µg/mL by diluting 10 mg of the compound in PBS. The solution was vortexed thoroughly and aliquoted into Eppendorf tubes (50 µL, *n* = 3). Each sample was labeled according to the designated incubation times (0, 6, 12, 24, and 48 h) and placed in a water bath (Shaking Water Bath, JS Research, Gong-Ju, Republic of Korea) maintained at 32 °C and 60 rpm. At the respective time points, samples were retrieved, followed by the addition of 200 µL of methanol (MeOH). After vortexing, the mixtures were filtered and subsequently subjected to analysis.

2.5. Analytical Method Validation

Calibration Standards and Quality Control (QC) Samples: Accuracy and Precision

Matrix-matched calibration standards were prepared separately for each matrix. For SKIN, calibration levels were 0.1, 0.5, 1, 3, 5, and 10 µg/mL, prepared by spiking 5 µL of

working standard into 95 μL of SKIN blank (final 100 μL). For WASH, stratum corneum (SC), and receptor fluid (RF), calibration levels were 0.2, 0.5, 1, 3, 5, and 10 $\mu\text{g}/\text{mL}$, prepared identically (5 μL standard + 95 μL of the corresponding matrix blank). Calibration curves were constructed for each matrix. Each calibration solution was subsequently diluted 10-fold, yielding final concentrations of 0.01 or 0.02, 0.05, 0.1, 0.3, 0.5, and 1 $\mu\text{g}/\text{mL}$. Quality control (QC) samples were prepared at four concentration levels: the lower limit of quantification (LLOQ, 0.1 or 0.2 $\mu\text{g}/\text{mL}$), low QC (LOQ, 0.6 $\mu\text{g}/\text{mL}$), medium QC (MOQ, 4 $\mu\text{g}/\text{mL}$), and high QC (HOQ, 8 $\mu\text{g}/\text{mL}$). All prepared solutions were filtered prior to analysis, and stock solutions, calibration standards, and QC samples were stored at $-20\text{ }^{\circ}\text{C}$ until use. Method validation for R2 was evaluated based on linearity, sensitivity, selectivity, accuracy, and precision, using calibration curves and QC samples. Calibration curves were constructed by weighted ($1/x$) linear regression of the peak area ratios across the concentration range. Intra-day accuracy and precision were determined by analyzing triplicate samples in each matrix at 0.2, 0.6, 4, and 8 $\mu\text{g}/\text{mL}$ within a single day. The accuracy and precision were calculated using the following formula:

$$\text{Accuracy (\%)} = \text{measured concentration/nominal concentration} \times 100 \quad (1)$$

$$\text{Precision (\%)} = \text{standard deviation/nominal concentration} \times 100 \quad (2)$$

Inter-day accuracy and precision were assessed in the same manner over three consecutive days, using the same concentration levels. Accuracy and precision should be within 20% of the LLOQ concentration and within 15% of the LOQ, MOQ and HOQ according to the Ministry of Food and Drug Safety's bio-sampling method verification guidelines [35].

2.6. Preparation of Rat Skin

Sprague–Dawley (SD) rats (male, 8 weeks, $228 \pm 7\text{ g}$) were supplied by Samtako Co. (Osan, Republic of Korea). The experimental protocol was reviewed and approved by the Institutional Animal Care and Use Committee of Dankook University (IACUC approval number: 2018-014). For the *in vitro* experiments, the animals were sacrificed using CO_2 , and full-thickness dorsal skin was excised. The dorsal hair was removed using an electronic hair clipper (SM-129, JOAS, Namyangju, Republic of Korea) over an area of $7 \times 7\text{ cm}^2$. The subcutaneous fat was carefully removed from the excised skin using forceps. The prepared skin samples were stored at $-20\text{ }^{\circ}\text{C}$ until use in the diffusion experiments.

2.7. Franz Diffusion Cell and Sample Collection

This study followed the Korea Ministry of Food and Drug Safety (MFDS) guideline and OECD Guideline 428 for *in vitro* skin absorption method [36,37]. The Franz diffusion cell system was used for the dermal absorption test. The Franz diffusion cell system was composed of a Vision microette auto-sampler, circulating waterbath, stirring drive, stirring control, and autofill (Hanson, Los Angeles, CA, USA). The prepared skin was thawed at room temperature and hydrated in saline for 5 min. The skin was fixed between the donor and receptor chamber and checked to ensure there were no air bubbles between the receptor fluid (PBS) and the skin. Pre-weighed formulations were applied to the dosing area. Each formulation of 200 mg ($113\text{ mg}/\text{cm}^2$, $3.39\text{ mg}/\text{cm}^2$ of R2) was applied to 1.77 cm^2 of the donor chambers. Sampling times were set at 0, 1, 2, 4, 8, 12, and 24 h following dose application. After 24 h of dosing, the remaining formulation was collected using alcohol swabs (WAHS, BDTM Alcohol Swabs, Becton Dickinson Corp, Franklin Lakes, NJ, USA) to measure unabsorbed R2. The tape stripping method was used to remove the remaining formulation from the stratum corneum. The stratum corneum was removed by applying $2 \times 2\text{ cm}$ pieces of tape (Scotch™, 3M, Maplewood, MN, USA) sequentially

15 times. The SKIN process involves cutting the skin (epidermis and dermis) into eight pieces using surgical scissors and placing it in solvent to be extracted. The WASH, stratum corneum (SC), and SKIN used in each step were put in 20 mL PBS, sonicated for an hour, and stored at $-20\text{ }^{\circ}\text{C}$ for 24 h before analysis. All experiments were performed using skin obtained from three independent animals, with one diffusion cell per animal for each formulation ($n = 3$ technical replicates per formulation).

2.8. Statistical Analysis

All data are presented as the mean $\pm$ standard deviation (SD). Statistical calculations were performed using Excel 2019 (Microsoft for Windows) and GraphPad Prism software version 5.04 (San Diego, CA, USA).

3. Results

3.1. Stability Test

The stability test results confirmed that sink conditions for R2 were achieved under the planned skin permeation test conditions using phosphate-buffered saline (PBS) as the receptor fluid at $32 \pm 1\text{ }^{\circ}\text{C}$. No time-dependent change in concentration was detected at 0, 6, 12, 24, or 48 h, and no statistically significant differences were observed among the time points. R2 remained stable for 48 h in PBS under sink conditions (Figure 2). Accordingly, R2 was selected for the in vitro skin permeation experiment ($n = 3$).

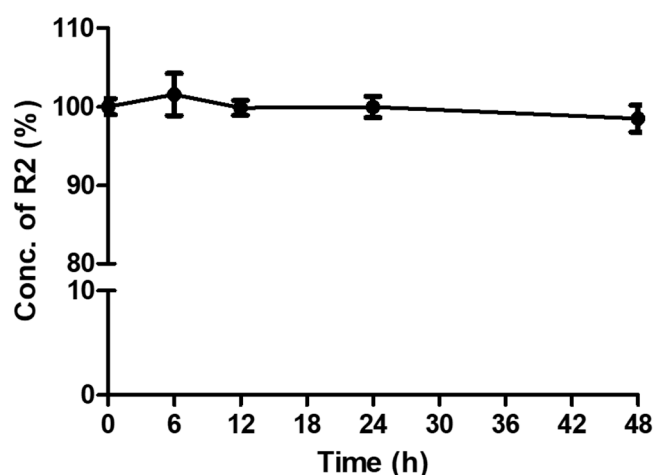


Figure 2. Stability of amaranth (R2) in phosphate-buffered saline (PBS).

3.2. Analytical Method Validation of R2

Calibration curves were constructed using seven concentration levels in accordance with the Korea (MFDS) (2013) guideline for bioanalytical method validation [35]. The retention time of R2 was 2.01 min, and no disturbing peaks were observed that interfered with the analysis (Figure 3). Following the pretreatment described above, HPLC analysis of R2-spiked samples yielded the calibration curve shown, which is representative, displaying excellent linearity ($r^2 > 0.9996$). Matrix-matched calibration curves showed good linearity over 0.1–10 $\mu\text{g/mL}$ for SKIN and 0.2–10 $\mu\text{g/mL}$ for WASH, SC, and receptor fluid (RF). Intra- and inter-day accuracy and precision were evaluated to determine the reliability of the current analytical method. Intra- and inter-day accuracy and precision were assessed at four QC levels (0.2, 0.6, 4.0, and 8.0 $\mu\text{g/mL}$ of R2) (Table 3). All results complied with MFDS (2013) bioanalytical validation criteria—accuracy within $\pm 20\%$ and coefficient of variation (CV) $\leq 20\%$ at the LLOQ (0.1 $\mu\text{g/mL}$ for WASH/SC/RF; 0.2 $\mu\text{g/mL}$ for SKIN) and within $\pm 15\%$ at the other levels. Matrix-matched performance was as follows: WASH, accuracy 95.5–99.8% with CV 0.6–5.5%; SC, 98.0–106.9% with CV 0.4–5.8%; SKIN,

100.5–104.4% with CV 0.5–3.9%; and RF, 99.4–102.9% with CV 0.3–5.8%. Collectively, these data confirm the reliability of the method for quantifying R2 [35].

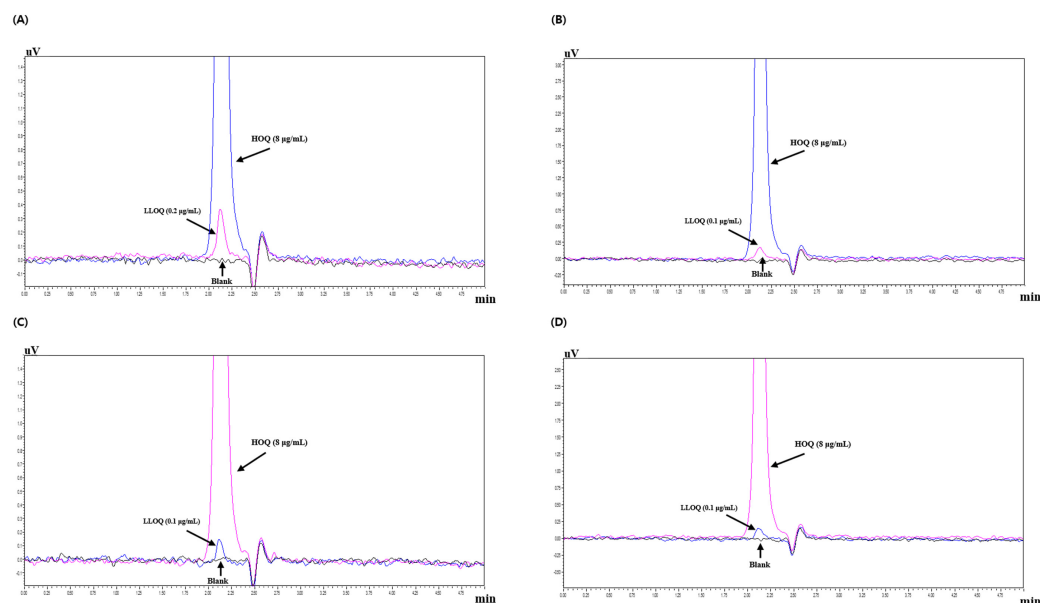


Figure 3. Representative chromatograms of Amaranth (R2) in the blank, LLOQ (0.1 or 0.2 µg/mL), and HOQ (8 µg/mL) samples across different matrices: (A) skin (dermis and epidermis), (B) receptor fluid (RF), (C) wash solution (remaining wash-out), and (D) stratum corneum (SC).

Table 3. The intra-day and inter-day accuracy and precision of amaranth (R2) analyses.

Sample ^a	Concentration (µg/mL)	Intra-Day			Inter-Day		
		Mean ± S.D.	Accuracy (%)	Precision (%)	Mean ± S.D.	Accuracy (%)	Precision (%)
SKIN (n = 3)	0.2	0.21 ± 0.01	104.4	3.9	0.20 ± 0.01	102.2	1.9
	0.6	0.61 ± 0.01	102.3	0.9	0.61 ± 0.01	101.4	0.9
	4.0	4.04 ± 0.08	100.9	2.0	4.02 ± 0.06	100.5	0.9
	8.0	8.05 ± 0.11	100.6	1.4	8.04 ± 0.14	100.5	0.5
WASH (n = 3)	0.1	0.10 ± 0.01	95.9	5.5	0.10 ± 0.01	98.7	3.2
	0.3	0.29 ± 0.01	95.5	2.9	0.29 ± 0.01	97.4	2.1
	4.0	3.93 ± 0.10	98.1	2.7	3.97 ± 0.07	99.3	1.2
	8.0	7.98 ± 0.14	99.8	1.8	7.97 ± 0.09	99.6	0.6
SC (n = 3)	0.1	0.11 ± 0.002	106.9	1.8	0.10 ± 0.01	98.0	5.8
	0.3	0.31 ± 0.003	103.3	1.1	0.29 ± 0.01	99.5	3.1
	4.0	4.00 ± 0.08	100.1	2.2	3.97 ± 0.05	100.5	0.4
	8.0	8.01 ± 0.05	100.1	0.6	7.97 ± 0.05	99.0	0.7
RF (n = 3)	0.1	0.10 ± 0.003	99.6	2.9	0.10 ± 0.004	102.9	5.8
	0.3	0.30 ± 0.01	100.5	2.8	0.31 ± 0.003	102.1	2.5
	4.0	4.03 ± 0.10	100.8	2.5	4.04 ± 0.06	101.1	1.9
	8.0	7.95 ± 0.02	99.4	0.3	8.00 ± 0.08	100.0	1.0

^a SKIN, Dermis and epidermis; WASH, Remaining wash-out; SC, Stratum Corneum; RF, Receptor fluid.

3.3. In Vitro Dermal Absorption of R2

After completion of the permeation study, residual R2 in each compartment was quantified. In the skin lotion formulation, recoveries were $76.1 \pm 5.9\%$ (WASH), $0.8 \pm 0.3\%$ (SC), $3.4 \pm 2.7\%$ (SKIN), and $0 \pm 0\%$ (RF), yielding a total recovery of $80.3 \pm 8.2\%$. In the cream formulation, the corresponding values were $91.0 \pm 19.6\%$ (WASH), $0.3 \pm 0.2\%$ (SC), $0 \pm 0\%$ (SKIN), and $0 \pm 0\%$ (RF), with a total recovery of $91.4 \pm 19.4\%$ (Table 4). The amounts

reaching the receptor fluid (RF) and the calculated permeation rates for each R2-containing formulation are summarized in Figure 4. The total dermal absorption of R2 was $3.4 \pm 2.7\%$ ($34.5 \pm 27.0 \mu\text{g}/\text{cm}^2$) for the skin lotion and $0 \pm 0\%$ ($0 \mu\text{g}/\text{cm}^2$) for the cream (Figure 4).

Table 4. In vitro dermal absorption *n* results for amaranth (R2).

	Skin Lotion (%) (Mean $\pm$ Standard Deviation)	Cream (%) (Mean $\pm$ Standard Deviation)
Wash (24 h)	76.1 ± 5.9	91.0 ± 19.6
¹ S.C	0.8 ± 0.3	0.3 ± 0.2
² Skin	3.4 ± 2.7	0
³ RF	0	0
⁴ Total dermal absorption	3.4 ± 2.7	0
Total recovery	80.3 ± 8.2	91.4 ± 19.4

¹ SC: stratum corneum. ² SKIN: dermis + epidermis. ³ RF: receptor fluid (PBS). ⁴ Total dermal absorption: dermis + epidermis + receptor fluid.

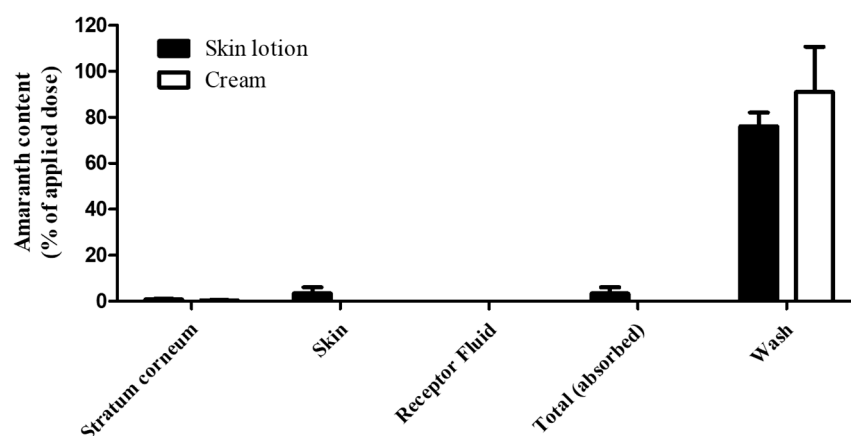


Figure 4. Deposition of amaranth in SKIN (dermis and epidermis), WASH (remaining wash-out), SC (stratum corneum) and RF (receptor fluid) after 24 h in vitro dermal application of skin lotion and cream formulations. Error bar showed standard deviation ($n = 3$).

4. Discussion

The skin absorption rate is a critical parameter in cosmetic risk assessment, as it directly determines the extent to which a cosmetic ingredient enters the body through the skin, thereby influencing systemic toxicity and the overall safety profile of the product. In this context, the present study was designed to generate essential input data for cosmetic risk assessment rather than to perform a complete risk assessment. First, the skin absorption rate is essential for estimating the systemic exposure dosage (SED), which serves as the foundation for cosmetic safety evaluations. Accurate SED values indicate how much of an ingredient may reach systemic circulation and target organs under realistic usage conditions [26]. Second, this parameter plays a pivotal role in establishing the margin of safety (MoS), a core concept in regulatory toxicology. By quantifying the absorbed fraction, regulators and researchers can confirm that human exposure remains well below levels associated with adverse effects. Finally, data derived from skin absorption studies are directly applied to MoS calculations for individual cosmetic ingredients, ensuring compliance with regulatory requirements and minimizing potential health risks to consumers. Collectively, these considerations underscore the indispensable role of skin absorption data in risk assessment frameworks for cosmetics.

Rat skin is commonly employed in percutaneous absorption studies owing to practical advantages such as ease of availability, small size, and low cost. However, it exhibits

markedly higher permeability compared to human skin, often by an order of magnitude or more. For example, mean in vitro permeation flux through rat skin has been reported to be approximately elevenfold greater than that of human skin across diverse physicochemical classes of compounds, and for certain lipophilic molecules, this difference can reach up to 50-fold [38,39]. One of the principal reasons for this discrepancy is the thinner stratum corneum in rats, which provides a weaker barrier relative to human skin [39,40]. Surface properties also differ, as rat skin tends to be hairier than human skin, although hairless strains are sometimes utilized to better approximate human permeation characteristics [39]. Given these differences, rat skin models are generally regarded as providing a “worst-case scenario” for human risk assessment. Consequently, dermal absorption data generated in rats require careful interpretation, and appropriate adjustment factors or safety margins should be applied when extrapolating to human exposure estimates [38–41].

In this study, rat skin was used instead of human skin in order to obtain dermal absorption data under more stringent conditions, thereby providing a conservative estimate of safety. As noted above, R2 has been prohibited in children’s confectionery and in oral formulations such as mouthwashes and oral wipes due to safety concerns [10]. In the United States, its use has been banned since 1976, and it is also prohibited as a food additive to strengthen consumer safety management [42].

This study demonstrated the dermal absorption characteristics of R2 using Franz diffusion cells under two representative cosmetic formulations (skin lotion and cream). Currently, no clear regulatory limits on the permissible concentration of R2 in cosmetics are established in Korea or the European Union. In particular, the European SCCS has prohibited its use as a hair dye ingredient; therefore, no reference concentration can be directly cited. The usage status of Amaranth (R2) in Korea was reviewed, and a test concentration of 1%—representing the maximum reported level of use in cosmetic products—was selected to provide a conservative exposure scenario. For this reason, two representative leave-on formulations reflecting realistic consumer exposure scenarios were selected, and a test concentration of 1% was applied. This level was chosen based on the general range of incorporation for tar dyes in cosmetic formulations, representing a conservative approach for safety assessment [43]. Importantly, the use of rat skin represents a conservative, worst-case experimental scenario due to its higher permeability compared with human skin. Therefore, the very low dermal absorption observed in this study further strengthens the conclusion that systemic exposure to R2 is expected to be minimal under realistic human use conditions. The results of the in vitro dermal absorption studies revealed distinct distribution patterns depending on the formulation. These formulation-dependent differences can be explained by the physicochemical characteristics of oil-in-water (O/W) and water-in-oil (W/O) emulsions. O/W formulations generally facilitate the release of hydrophilic, water-soluble compounds into the aqueous phase, enhancing their availability at the skin surface and promoting diffusion into the stratum corneum. In contrast, W/O formulations tend to retain hydrophilic ingredients within the internal aqueous phase and the occlusive lipid matrix, thereby limiting their release and reducing dermal penetration [44,45]. In the skin lotion, the majority of R2 remained on the skin surface and was removed by washing ($76.1 \pm 5.9\%$), while only a small fraction was retained in the stratum corneum ($0.8 \pm 0.3\%$) and viable skin layers ($3.4 \pm 2.7\%$). No R2 was detected in the receptor fluid. In the cream formulation, $91.0 \pm 19.6\%$ of the applied dose was removed by washing and $0.3 \pm 0.2\%$ was retained in the stratum corneum, whereas neither the viable skin layers nor the receptor fluid showed detectable levels. Accordingly, the final dermal absorption rates were determined to be $3.4 \pm 2.7\%$ for the skin lotion and 0% for the cream, with total recoveries of $80.3 \pm 8.2\%$ and $91.4 \pm 19.4\%$, respectively. These findings indicate that the vast majority of the applied dose remains on the skin surface and does not penetrate

into systemic circulation. The overall recovery ranged from 80 to 91%, and the minor losses observed are likely attributable to adsorption of the test substance to experimental materials, incomplete extraction from skin matrices, and unavoidable losses during sample handling and processing. The low dermal absorption of R2 can be further explained by its physicochemical properties. According to SCCS guidance (2023), compounds with a molecular weight above 500 Da, $\log P \leq -1$ or ≥ 4 , melting point > 200 °C, and topological polar surface area (TPSA) > 120 Å² are expected to exhibit limited skin penetration [32]. R2 possesses all these characteristics, with a molecular weight of 604.5 Da, $\log P$ of -5.13 , melting point > 300 °C, and TPSA of 242 Å². These properties are consistent with the observed minimal absorption in our experiments, supporting the conclusion that systemic exposure to R2 via dermal routes is negligible under realistic cosmetic use conditions.

Taken together, the present findings highlight that dermal absorption of R2 is minimal under cosmetic use conditions, consistent with predictions based on its physicochemical profile. Although the use of R2 as a cosmetic colorant remains restricted in several jurisdictions, the conservative test design and low absorption rates observed here provide valuable data for risk assessment.

5. Conclusions

In this study, the dermal absorption of Amaranth (R2), a synthetic azo dye used as a cosmetic colorant, was evaluated using an in vitro Franz diffusion cell system with rat skin under two representative formulations (skin lotion and cream) at a concentration of 1%. A validated HPLC method ensured reliable quantification, and the results indicated minimal absorption. In the lotion formulation, most of the applied dose was removed by washing, with only $3.4 \pm 2.7\%$ (34.5 ± 27.0 µg/cm²) detected in skin and none in the receptor fluid. In the cream formulation, absorption was negligible (0%, 0 µg/cm²), with nearly all of the substance remaining on the skin surface. Collectively, to the best of our knowledge, this study provides the first validated dermal absorption data for R2 and offers valuable information for future safety assessments and regulatory considerations. However, these findings should be interpreted in light of certain limitations, including the use of rat skin as a surrogate for human skin and the inherent constraints of the in vitro Franz diffusion system. Future studies should confirm these findings using stratum corneum-mimetic membranes (e.g., Strat-M®) and human-relevant skin models, while evaluating concentration-dependent dermal absorption across a broader range of cosmetic formulations with diverse excipient systems to further strengthen the translatability and generalizability of the present results.

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